

3G_4G Digital Voice Interface

Application Note

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1 Applicability Table

Table 1: Applicability Table

Products	SW Version
HE910 Family	
HE910	12.00.xx3
HE910-GL	12.00.xx4
HE910-EUR	12.00.xx4
HE910-NAR	12.00.xx4
HE910-EUG	12.00.xx4
HE910-NAG	12.00.xx4
UE/UL Family	
UE910-EUR	12.00.xx4
UE910-NAR	12.00.xx4
UL865-EUR	12.00.xx4
UL865-NAR	12.00.xx4
UL865-BR	12.00.xx4
LE910 Family	
LE910 Cat1	20.00.xxx
LE910 V2	20.00.xxx
MEx10G1 Family	
ME910 WWV	37.00.xx4
ME310 WWV	37.00.xx4
ML865 WWV	37.00.xx4
LE910S1 Family	
LE910S1-xxx	39.00.xxx
LE910R1 Family	
LE910R1-xxx	44.00.xxx



2 Introduction

2.1 Scope

This document provides a comprehensive guideline for setting up and using the Digital Voice Interface implemented on Telit's module families as listed in the Applicability Table.

The document covers the configurations of the Digital Voice Interface such as sampling frequency, bit depth, audio formats, and so on.

Additionally, the document demonstrates specific configurations involving a commonly used Audio Codec that is connected to the Telit module. The communication between the module and the Audio Codec is established using I2S (Inter-Integrated Circuit Serial) and I2C (Inter-Integrated Circuit) buses. However, detailed information regarding the specific hardware characteristics of these buses is beyond the scope of this document.

2.2 Audience

This document is intended for users who need to develop applications dealing with signal voice in digital format.

2.3 Contact Information, Support

For technical support and general questions, e-mail:

- TS-EMEA@telit.com
- TS-AMERICAS@telit.com
- TS-APAC@telit.com
- TS-SRD@telit.com
- TS-ONEEDGE@telit.com

Alternatively, use: <https://www.telit.com/contact-us>

Product information and technical documents are accessible 24/7 on our website:
<https://www.telit.com>

2.4 Conventions

Note: Provide advice and suggestions that may be useful when integrating the module.

Danger: This information MUST be followed, or catastrophic equipment failure or personal injury may occur.

ESD Risk: Notifies the user to take proper grounding precautions before handling the product.

Warning: Alerts the user on important steps about the module integration.

All dates are in ISO 8601 format, that is YYYY-MM-DD.

2.5 Terms and conditions

Refer to <https://www.telit.com/hardware-terms-conditions/>.

2.6 Disclaimer

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3 Overview

This chapter provide an overview of Telit's Digital Voice Interface (DVI) and discuss its usage and application scenarios.

Telit's DVI is a feature that enables the handling of digital voice signals within the module. It allows for the integration of voice communication capabilities in various applications.

Figure 1, illustrates a typical application scenario.

Initially, the module receives a downlink signal from the radio access network. The module's dedicated software captures incoming voice packets and directs them to the Digital Voice Interface. The Audio Codec converts digital voice samples into an analog signal that is amplified and then sent to the speaker for audio output.

Conversely, in the opposite direction, the microphone captures acoustic speech, and sampled by the Audio Codec in the digital domain. The audio samples are directed from the codec through the Digital Voice Interface to the module. The module collects and encodes the received voice stream.

Finally, the encoded voice stream is transmitted through the uplink channel back to the network.

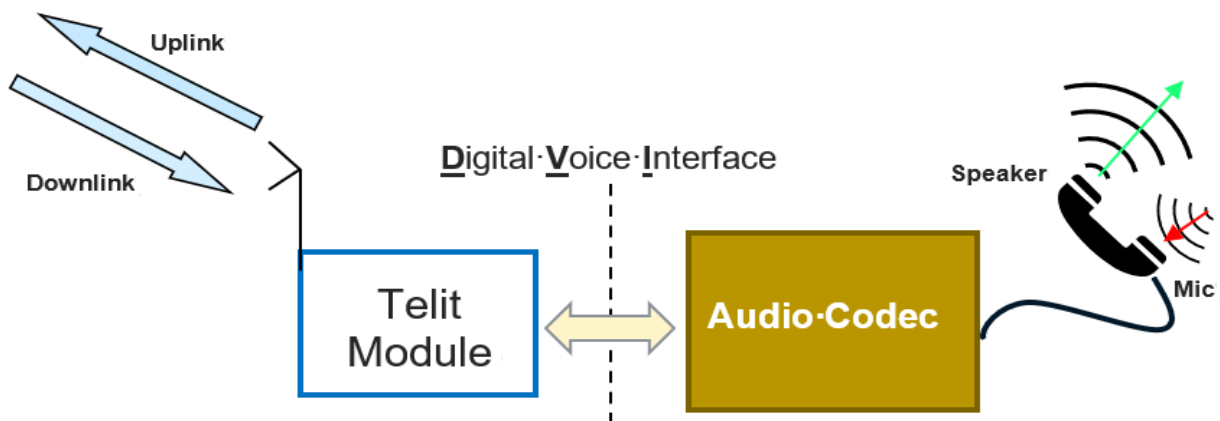


Figure 1:Example of Digital Voice Interface Use

Note: Telit's module supports also the Echo canceller functionality, which is beyond the scope of this document. Please refer to HE910/UE910/UL865 AT Commands Reference Guide for the specific AT commands.

4 Technical Description

The DVI interface provided by the Telit's module is based on the I2S (Inter-Integrated Circuit Serial) bus protocol. An overview of the standard I2S serial bus is described under section 5.1 Normal Mode (I2S).

Considering the DVI from the electrical point of view, four wires carry two pairs of digital signals referred to a common ground.

Table 2, summarizes the DVI signals.

Table 2: DVI Signals

Headline	Description	Note
Clock	DVI_CLK	Data Clock
Word Alignment	DVI_WAO	Frame Synchronism
Serial audio data input	DVI_RX	Received Data
Serial audio data output	DVI_TX	Transmitted Data

For detailed information on the electrical characteristics and pin-out of the signals pin-out of the signals, refer to documents [1], [4], and [5].

The Clock and Word Alignment Output (WAO) signals are shared for synchronization purposes, while two I2S data lines carry the received and transmitted audio signals.

There are two possible configurations based on the source of the Clock and WAO signals:

- **Master Mode (Figure 2):** When the module operates as the master, it generates the Clock and WAO signals internally.



Figure 2: Master Configuration

- **Slave Mode (Figure 3):** On the other hand, when the module is in Slave mode, the Clock and WAO signals are generated by an external codec device.

Both of these configurations are available using the AT#DVI command provided by Telit*.

Note: *LE910S1 and LE910R1 families support only master mode.



Figure 3: Slave Configuration

Additionally, the AT#DVIEXT command offers additional settings including sampling frequency, sample width, and communication protocol.

The tables in the following sub-sections summarize the parameters of AT#DVI and AT#DVIEXT commands.

For a more general overview of AT commands and syntax details, refer to documents [3].

4.1 AT#DVI

AT#DVI command enables/disables the DVI interface, selects the DVI port, and sets the module in Master or Slave configuration. The following table shows the possible value for each parameter.

AT#DVI =<mode>,<dviport>,<clockmode>

Table 3: DVI Configuration via AT#DVI command

<MODE>	<DVIPORT>	<CLOCKMODE>
0 → disable DVI interface ¹	1 → reserved	0 → DVI slave ^{4 5}
1 → enable DVI interface ²	2 → select DVI port ¹	1 → DVI master
2 → reserved ³		

Note:

¹ Default setting for UE910 products

² Default setting for HE901 UL865 and MEx10G1 family products

³ On MEx10G1 family factory settings are used, ignoring all the remaining parameters of AT#DVI and AT#DVIEXT

⁴ Not supported on MEx10G1 family using PCM mode

⁵ Not supported on LE910S1 and LE910R1 families.

4.2 AT#DVIEXT

AT#DVIEXT command sets the DVI mode in either Normal I2S or Burst PCM:

- In Normal I2S mode, WAO signal defines the left and right audio channel.
- In Burst PCM mode, WAO signal defines the beginning of the audio frame.

The following table shows the possible value for each parameter.

AT#DVIEXT = <CONFIG>,<SAMPLERATE>,<SAMPLEWIDTH>,<AUDIOMODE>,<EDGE>

Table 4: DVI Audio Format configuration via AT#DVIEXT command

<CONFIG>	<SAMPLERATE>	<SAMPLEWIDTH>	<AUDIOMODE>	<EDGE>
0 → Burst PCM	0 → 8 kHz	0 → 16 bits 1 → 18 bits ^{2,3}	0 → Mono ³	0 → FALLING on TX RISING on RX
1 → I2S ¹	1 → 16 kHz ²	2 → 20 bits ^{2,3} 3 → 24 bits ^{2,3} 4 → 32 bits ^{2,3}	1 → Dual Mono	1 → FALLING on RX RISING on TX ^{2,4}

Note:

¹Default setting

²Not supported on MEx10G1 family

³Not supported on LE910S1 and LE910R1 families

⁴Reserved on I2S mode

By setting the edge parameter to 0, the falling edge of the clock is used to shift out the next data to transmit. The received data bit is captured on the rising edge of the clock.

Otherwise, the rising edge of the clock is used to shift out the next data to transmit. The received data bit is captured on the falling edge of the clock.



5 DVI AT Commands

This chapter provides a detailed list of configurations available in both Master and Slave modes. The following examples are performed using the hardware setup shown in Figure 4.

In this setup, the I2C bus standard is used to program the MAX9867 Codec . A set of GPIO on the Telit's module are configured for this purpose. y issuing appropriate AT commands, the user can program the codec. As mentioned earlier, the DVI bus provides the voice connection between the two devices.

The following examples use the MAX9867 Codec. For a schematic reference design of this codec, refer to section 6.3 Recommended Hardware Design. In general, the user can use any codec compliant with the technical requirements of the Telit's modules.

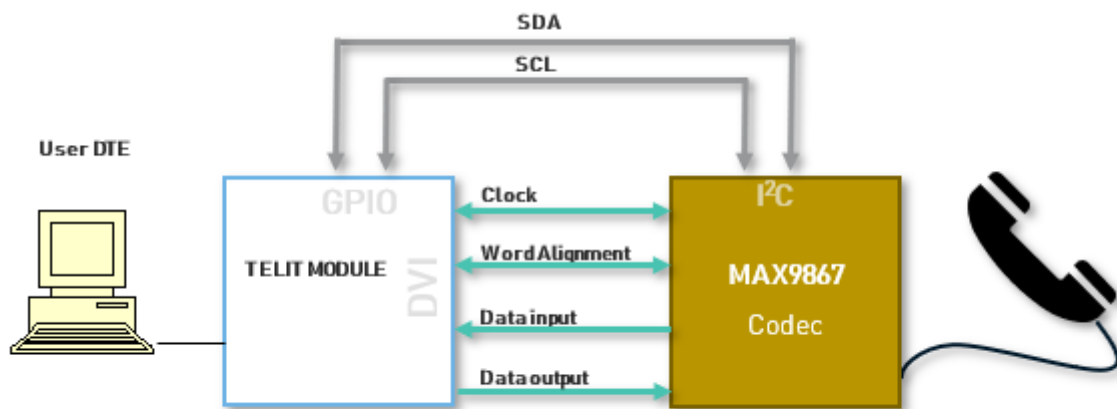


Figure 4: Telit's module/Codec Connections

The setting examples are organized as shown in the below figure.

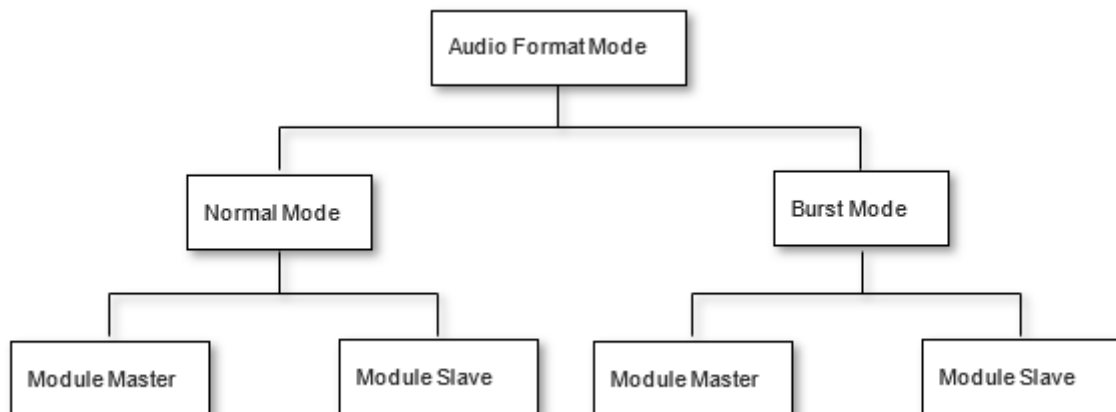


Figure 5: DVI Configurations

5.1 Normal Mode (I2S) Module is Master

The below figure illustrates a timing diagram that refers to the module in the role of master. In this configuration, the module generates the WAO and CLK signals. . The WAO signal defines the frame for the two audio channels: left and right.

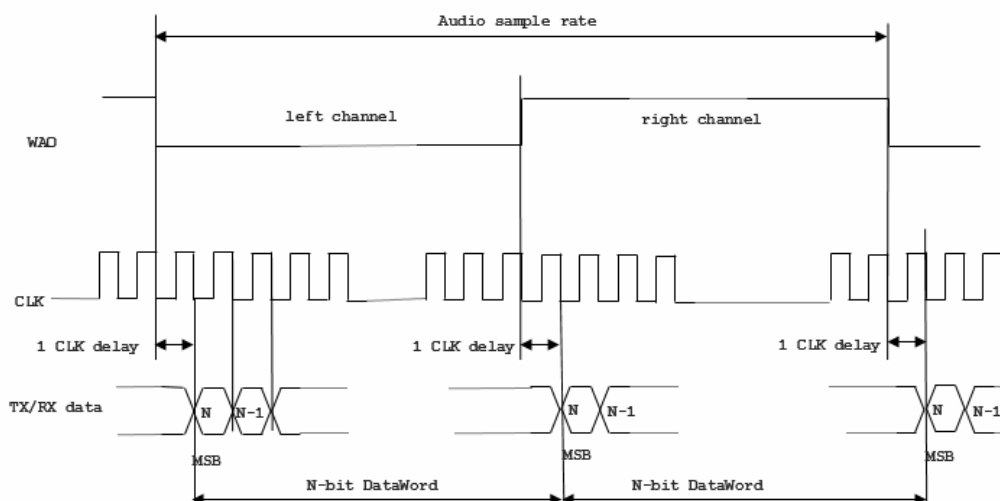


Figure 6:Module is Master/Normal mode/N bits per sample/Dual Mono

When the module is Master, the Bit Clock Frequency (CLK) is calculated using the following expression:

$$\text{Clock Frequency} = \text{Bit Depth} \times \text{Channel Number} \times \text{Sampling Rate}$$

The following table presents the resulting clock frequencies using all the possible settings.

Table 5: Bit Clock Frequency generated by the module in Master/Normal Mode

Bit Depth	Audio Channels	Clock freq. [kHz] Sampling rate @ 8kHz	Clock freq. [kHz] Sampling rate @ 16kHz
16	2	256	512
18	2	384*	768
20	2	384*	768
24	2	384*	768
32	2	512	1024

Note:

*The module generates 384 or 768 kHz also when the audio sample has 16 or 20 bits. In these configurations, only 16 or 20 bits are taken in consideration, and all other bits must be discarded.

The lists of AT commands used to set the module in Master Normal (I2S) are marked in gray. The MAX9867 Codec configurations are included in accordance with the module settings. The purpose and meaning of each setting are fully described to ensure clarity and ease of use.

First, configure the Telit module in Master Normal (I2S) mode.

```
AT#DVI=1,2,1
```

```
OK
```

Set the module as Master (factory setting)
Use DVI port 2 (mandatory)
Enable DVI interface

```
AT#DVIEXT=1,0,0,1,0
```

```
OK
```

The falling edge of the clock is used to shift out the next data to transmit, the received data bit is captured on the rising edge of the clock. (factory setting)
Dual Mono, the same Data Word is transmitted on both audio channels (factory setting)
16 bits per sample (factory setting)
Sample rate 8 kHz (factory setting)
Normal Mode (factory setting)

In addition, it is also necessary to verify whether the external codec configuration is Slave Normal (I2S) Mode.

Warning: The following examples are specific to the MAX9867 Codec using the I2C communication protocol. For other products, refer to the respective manufacturers datasheet.

GPIO number used as SDA, refer to [3]
GPIO number used as SCL, refer to [3]
Device address on I2C, refer to [2]
Register address from which start the writing, refer to [2]
Number of bytes to write

```
AT#I2CWR=X,Y,30,4,19
```

```
>00109000100A330000330C0C09092424400060
```

```
OK
```

MAX9867 specific settings, please refer to [2] and to suggested hardware design described at 6.3 Recommended Hardware Design

```
AT#I2CWR=X,Y,30,17,1
```

```
>8A
```

```
OK
```

Same as above

Figure 7, illustrates a timing diagram captured by a logic analyzer, using the previously described settings. The CLK at 256 kHz and WAO signals are generated by the module.



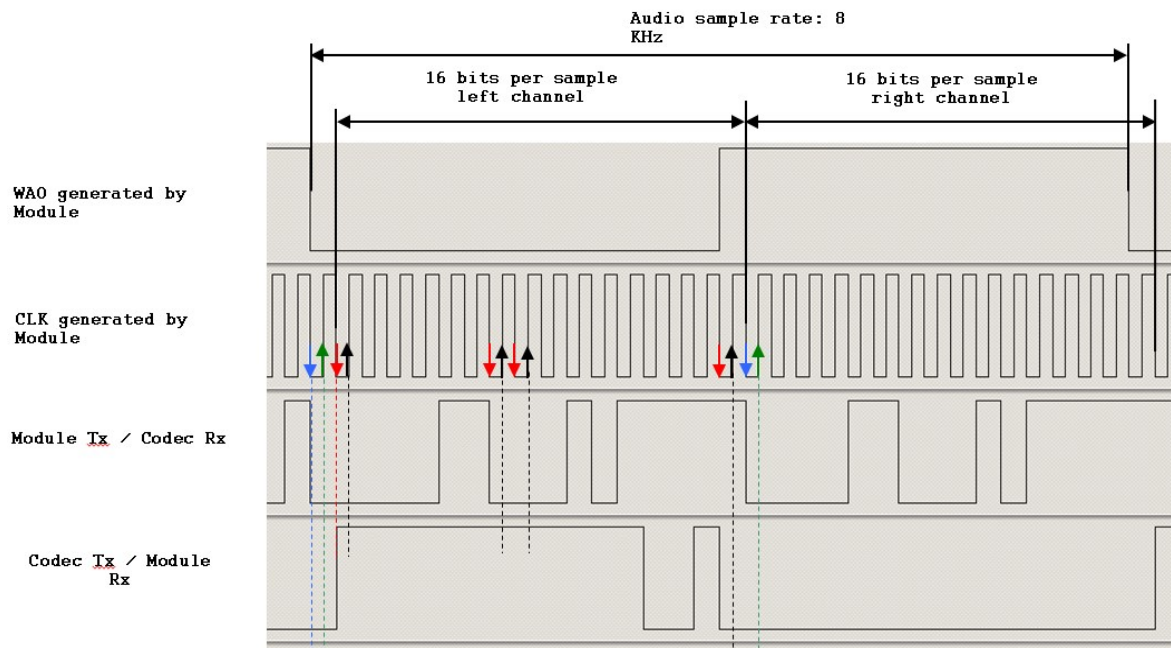


Figure 7: Module is Master/Normal Mode/16 bits per sample/Dual Mono/<edge>=0

Left channel:

↓ : Data transitions occur on the falling edge of the CLK

↑ : Data are latched on the rising edge of the CLK

Right channel:

↓ : Data transitions occur on the falling edge of the CLK

↑ : Data are latched on the rising edge of the CLK

Module is Slave

The list of AT commands used to set the module in Slave Normal (I2S) Mode are marked in gray. Mode and codec configuration are included in accordance with the module setting. The meaning of each setting is fully described for added convenience.

First, configure the Telit module in Slave Normal (I2S) Mode.

AT#DVI=1,2,0

OK

Set the module as Slave (factory setting)
Use DVI port 2 (mandatory)
Enable DVI interface

AT#DVIEXT=1,0,3,1,0

OK

The falling edge of the clock is used to shift out the next data to transmit, the received data bit is captured on the rising edge of the clock. (factory setting)
Dual Mono, the same Data Word is transmitted on both audio channels (factory setting)
24 bits per sample
Sample rate 8 kHz (factory setting)
Normal Mode (factory setting)

In addition, it is also necessary to verify whether the external codec configuration is Master Normal (I2S) Mode.

Warning: The following examples are specific to MAX9867 Codec using the I2C communication protocol. For other products, refer to the manufacturers data sheet.



Configure the codec in Master Normal (I2S) Mode.

GPIO number used as SDA, refer to [3]
GPIO number used as SCL, refer to [3]
Device address on I2C, refer to [2]
Register address from which start the writing, refer to [2]
Number of bytes to write

```
AT#I2CWR=X,Y,30,4,19
```

```
>001010009002330000330C0C09092424400060
```

```
OK
```

MAX9867 specific settings, please refer to [2] and to suggested hardware design described at 6.3 Recommended Hardware Design

```
AT#I2CWR=X,Y,30,17,1
```

```
>8A
```

```
OK
```

Same as above

Note: In this example the MAX9867 codec is in Master configuration and generates a clock equal to 384 kHz. On the module the selected number of bits per sample is 24.

Figure 8, illustrates a timing diagram captured by a logic analyzer, using the previously described settings. The CLK at 384 kHz and WAO signals are generated by the codec.

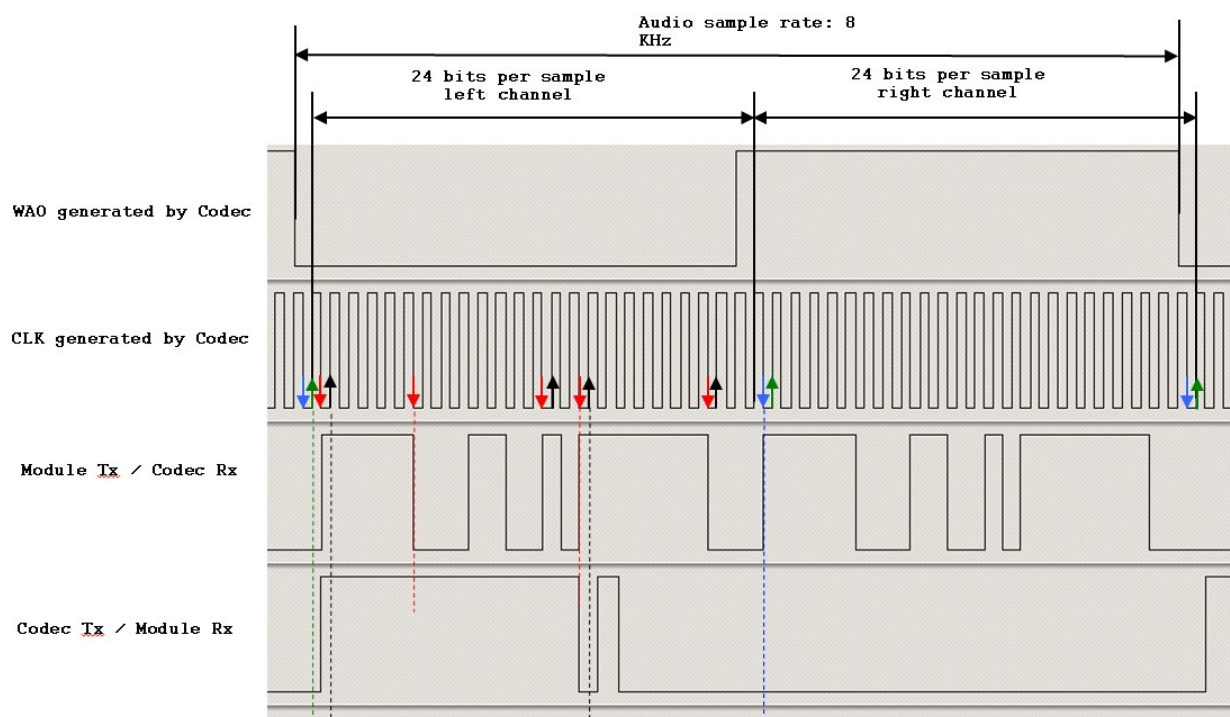


Figure 8: Module is Slave/Normal Mode/24 bits per sample/Dual Mono/<edge>=0

Left channel:

↓ : Data transitions occur on the falling edge of the CLK

↑ : Data are latched on the rising edge of the CLK

Right channel:

↓ : Data transitions occur on the falling edge of the CLK

↑ : Data are latched on the rising edge of the CLK

5.2 Burst Mode (PCM)

Module is Master

The list of AT commands used to set the module in Master using burst (PCM) mode are marked in gray. Mode, and codec configuration are included in accordance with the module setting. The meaning of each setting is fully described for added convenience.

First, we configure the Telit's module in Master Normal (I2S) Mode.

AT#DVI=1,2,1

OK

Set the module as Master (factory setting)
Use DVI port 2 (mandatory)
Enable DVI interface

AT#DVIEXT=0,0,0,0,1

OK

The rising edge of the clock is used to shift out the next data to transmit, the received data bit is captured on the falling edge of the clock
Mono Mode
16 bits per sample (factory setting)
Sample rate 8 kHz (factory setting)
Burst Mode PCM



Configure the codec in Slave Normal (PCM) Mode.

Warning: The following examples are specific to the MAX9867 Codec using the I2C communication protocol. For other products, please refer to the manufacturers data sheet.

GPIO number used as SDA, refer to [3]

GPIO number used as SCL, refer to [3]

Device address on I2C, refer to [2]

Register address from which start the writing, refer to [2]

Number of bytes to write

```
AT#I2CWR=X,Y,30,4,19
>00109000140A330000330C0C09092424400060
OK
```

MAX9867 specific settings, please refer to [2] and to suggested hardware design described at 6.3 Recommended Hardware Design

```
AT#I2CWR=X,Y,30,17,1
>8A
OK
```

Same as above

Module is Slave

Figure 9, illustrates a timing diagram that refers to the codec in master configuration. In this case, the WAO and CLK signals are generated by the codec.

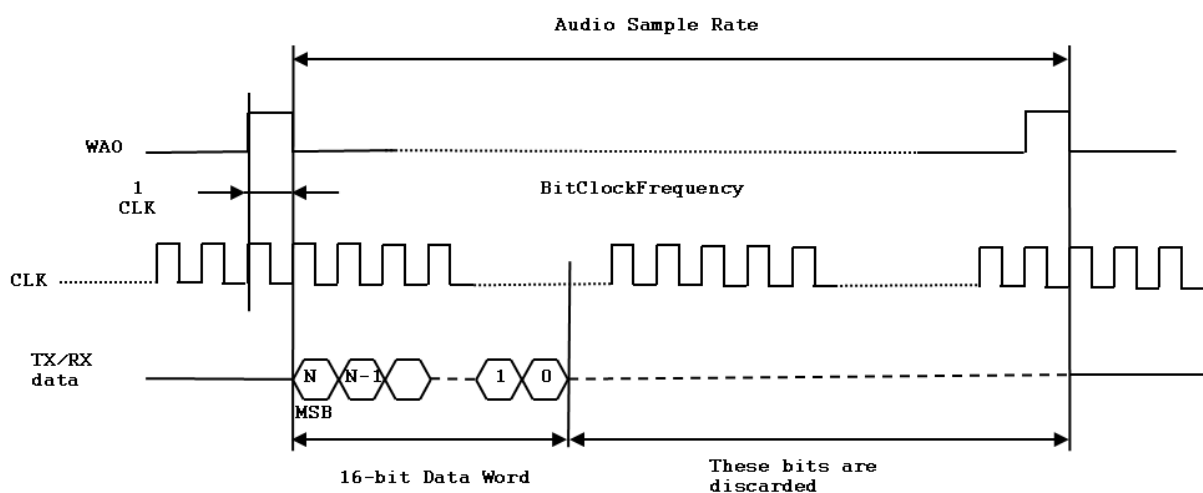


Figure 9: Module is Slave/Burst Mode/N bits per Sample/Mono Mode

Here, are the lists of AT commands used to set the module in Slave Burst (PCM) Mode and to configure the codec in accordance with the current module setting. As a convenience, the meaning of each setting is fully described.

Configure the module in Slave Burst (PCM) Mode.

First, we configure the Telit's module in Slave Burst (PCM) Mode.

```
AT#DVI=1,2,0
```

```
OK
```

Set the module as Slave (factory setting)
Use DVI port 2 (mandatory)
Enable DVI interface

```
AT#DVIEXT=0,0,0,0,1
```

```
OK
```

The rising edge of the clock is used to shift out the next data to transmit, the received data bit is captured on the falling edge of the clock
Mono Mode
16 bits per sample (factory setting)
Sample rate 8 kHz (factory setting)
Burst Mode PCM

Configure the codec in Master Burst (PCM) Mode.

Warning: The following examples are specific to MAX9867 Codec using the I2C communication protocol. For any other product please refer to the manufacturers data sheet.

GPIO number used as SDA, refer to [3]
GPIO number used as SCL, refer to [3]
Device address on I2C, refer to [2]
Register address from which start the writing, refer to [2]
Number of bytes to write

```
AT#I2CWR=X,Y,30,4,19
```

```
>00101000A40A330000330C0C09092424400060
```

```
OK
```

MAX9867 specific settings, please refer to [2] and to suggested hardware design described at 6.3 Recommended Hardware Design

```
AT#I2CWR=X,Y,30,17,1
```

```
>8A
```

```
OK
```

Same as above



Figure 10, illustrates a timing diagram, captured by a logic analyzer, using the previously described module/codec setting. The CLK at 384 kHz and WAO signals are generated by the codec.

↑ : Data transitions occur on the rising edge of the CLK

↓ : Data are latched on the falling edge of the CLK

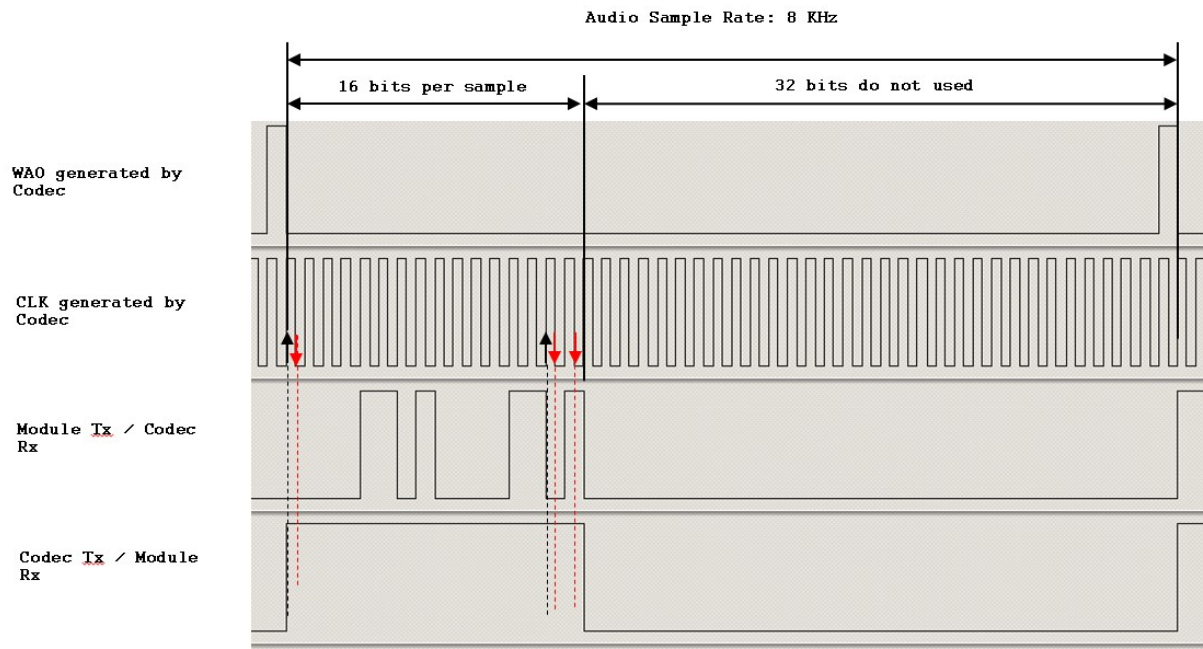


Figure 10: Module is Slave/Burst Mode/16 bits per Sample/Mono Mode/<edge>=1



6 Annex

6.1 I2S Overview

This section provides a brief description of the standard I2S bus which has been suitably modified for the DVI interface implemented on Telit's modules.

The standard I2S is an electrical serial bus designed for connecting digital audio devices. This popular serial bus was developed by Philips® in 1986 as a 3-wire bus for interfacing with audio chips, such as codecs. It is a simple data interface, without any form of addressing or device selection.

Referring to the figure below the I2S design handles audio data separately from clock signals. On an I2S bus, there is only one bus master and one transmitter.

In high-quality audio applications involving a Codec, the Codec typically acts as the master providing precise control over the I2S bus clock.

The I2S bus design consists of the following serial bus lines:

- SD: Serial Data
- WS: Word Select
- SCK: Serial Clock

The I2S bus carries two channels (left and right) 8 bit long, which are typically used to carry stereo audio data streams. The data alternates between the left and right channels, as controlled by the word select signal driven by the bus master.

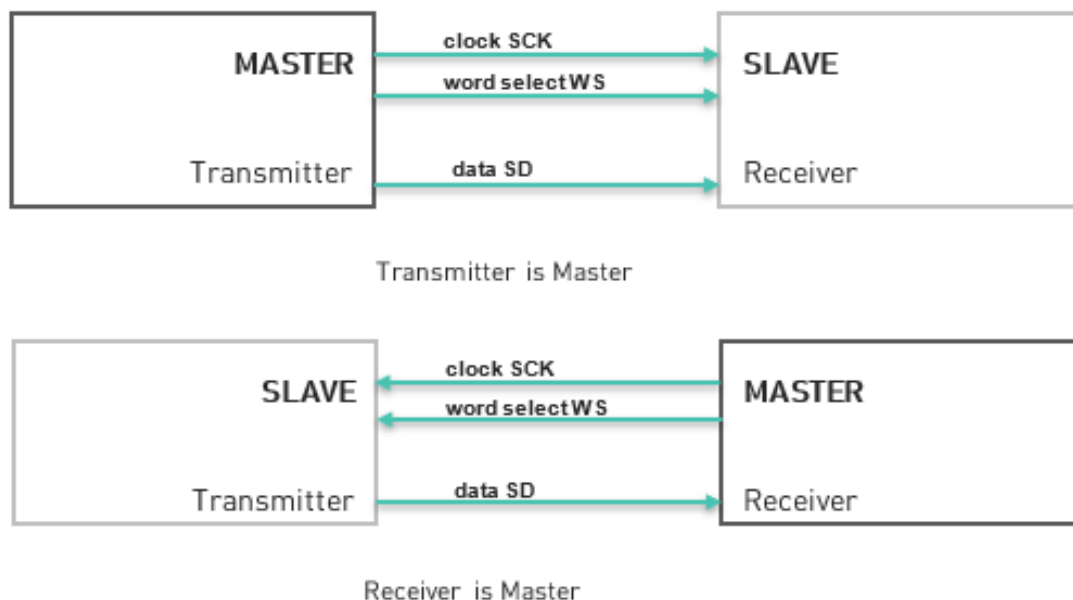


Figure 11: I2S Bus Configurations

6.2 PCM protocol (DSP mode)

The DSP/PCM mode is an extension of the standard I2S protocol. One of the purposes of the I2S DSP/PCM Mode (A/B) is to interface with a bus, which includes various external devices. The main difference between the DSP/PCM mode and the standard I2S mode is that the frame clock does not represent a different channel when high or low but is a bit-

wide pulse that marks the start of a frame. Data is aligned such that the left channel data is immediately followed by right channel data. Zero padding is filled in for the remaining bits. The data and frame clock may be configured to clock in on the rising or falling edge of Bit Clock.

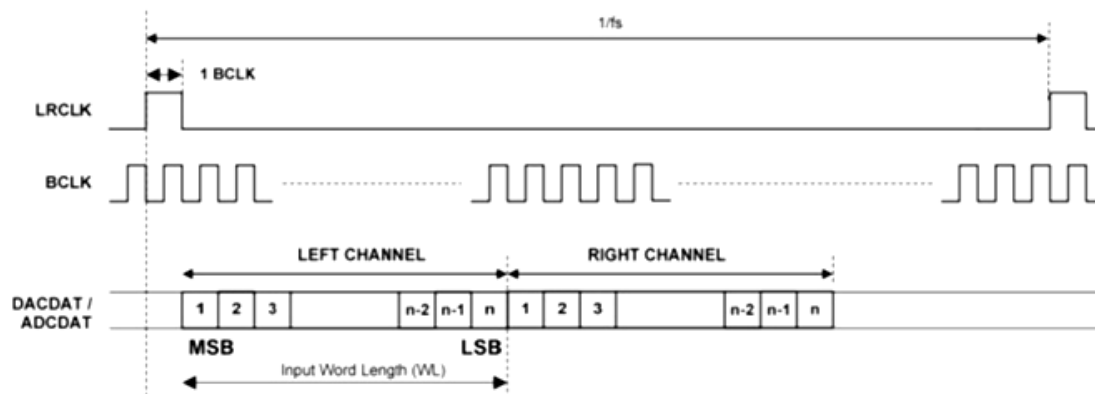


Figure 12: DSP/PCM Mode audio interface (mode B, LRP=0)

In DSP/PCM mode, the left channel MSB is available on either the first (Mode B) or second (Mode A) rising edge of BCLK (selectable by LRP) following a rising edge of LRC. Right channel data immediately follows left channel data. Depending on word length, BCLK frequency and sample rate, there may be unused BCLK cycles between the least significant bit (LSB) of the right channel data and the next sample.

Mode A and B are shown respectively in Figure 12 and Figure 13.

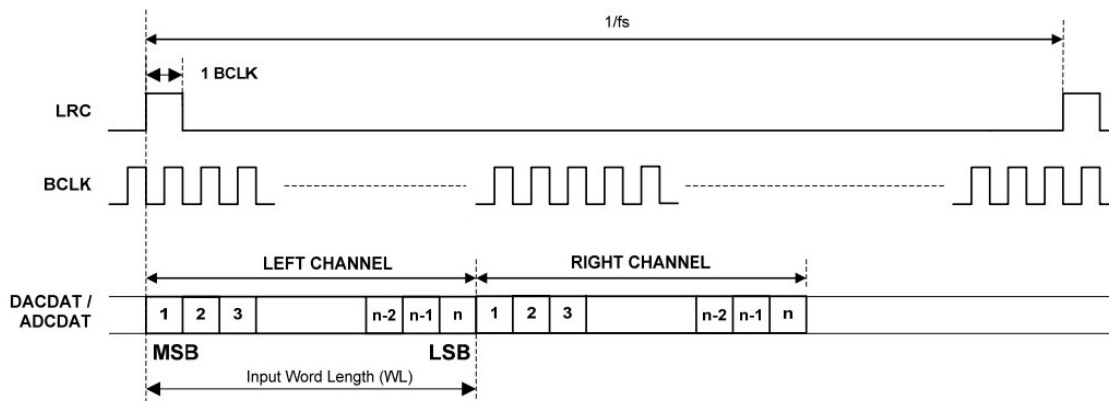


Figure 13: DSP/PCM Mode audio interface (mode B, LRP=0)

Operating the digital audio interface in DSP mode allows ease of use for supporting the various sample rates and word lengths.

The only requirement is that all data is transferred within the correct number of BCLK cycles to suit the chosen word length.

6.3 Recommended Hardware Design

We propose a design based on the MAX9867 audio codec (Figure 14), because it's simple and has a PLL onboard that allows the use of a Master Clock with a frequency ranging from 10 to 60 MHz. The frequency of 12.288 MHz was chosen as it is a commonly used frequency, but developers have the flexibility to use a different clock frequency that may already be available in their application.

Note: Changing the frequency of the Master Clock requires to modify accordingly the AT#I2CWR configuration commands described in section 4 of MAX9867 Ultra-Low Power Stereo Audio Codec document. To minimize the electrical interference and power consumption, the Master Clock is not provided by the Telit's modules described here (see Applicability Table). Where necessary, the Master Clock must be generated locally by a dedicated electrical device, for example, X101(Figure 14).

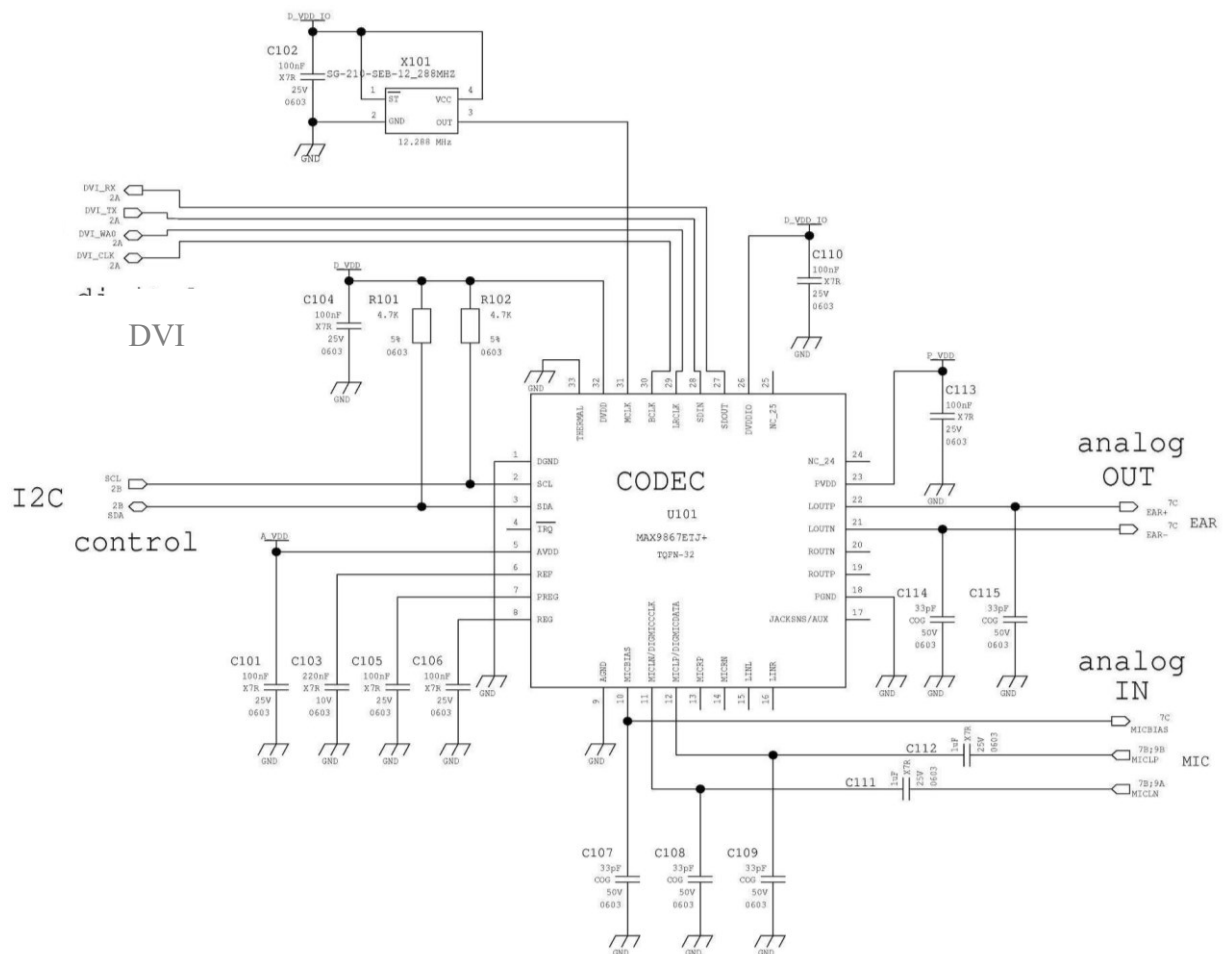


Figure 14: Schematic for Reference Design

Warning: Master Clock signal for the audio CODEC chipset is not provided by the Telit's module and it must be generated locally by a dedicated electrical device.

7 Acronyms and Abbreviations

Table 6: Acronyms and Abbreviations

Acronym	Definition
CLK	Clock
Codec	Coder-Decoder
DSP	Digital Signal Processing
DVI	Digital Voice Interface
I2S	Inter-IC Sound
PCM	Pulse Code Modulation
WAO	Word Alignment Output

8 Related Documents

Refer to <https://dz.telit.com/> for current documentation and downloads.

Table 7: Related Documents

S.no	Book Code	Document Title
1	1VV0300925	HE910 Hardware User Guide
2	MAXIM	MAX9867 Ultra-Low Power Stereo Audio Codec
3	80378ST10091A	HE910/UE910/UL865 AT Commands Reference Guide
4	1VV0301012	UE910 Hardware User Guide
5	1VV0301050	UL865 Hardware User Guide
6	80672ST11051A	LE910S1 AT Commands Reference Guide
7	80690ST11099A	LE910R1 AT Commands Reference Guide

9 Document History

Table 8: Document History

Revision	Date	Changes
13	2023-07-11	Added LE910S1 and LE910R1 support (in Applicability table, Related documents and AT#DVI, AT#DVIEXT footnotes) Updated template Editorial changes
12	2021-09-16	Changed template, various cosmetic modifications Added ME910 support Added BURST PCM configuration examples
11	2021-04-02	Added table DVI support, added ME910 family in Applicability table
10	2019-10-22	Added HE910-EUG, HE910-NAG and UL865-BR in Applicability table
9	2018-02-08	Changed the document title from "HE/UE910, UL865 Digital Voice Interface" to "3G/4G Digital Voice Interface".
8		
7	2017-02-13	2017 Template applied
6	2014-04-16	The note about the Echo canceller has been added in chapter 2. The chapters numbering/naming has been reorganized.
5	2014-03-17	In the Applicability Table have been corrected the wrong products, turning them into UL865-EUR / 12.00.xx4 and UL865-NAR / 12.00.xx4
4	2013-09-30	The previous document title "HE/UE910 Digital Voice Interface" has been changed in "HE/UE910, UL865 Digital Voice Interface".
3	2013-06-20	The previous document title "HE910 Family Digital Voice Interface" has been changed in HE/UE910 Digital Voice Interface. In accordance with the new title, the Applicability Table has been updated
2	2013-05-31	Updated all the screenshots of the timing diagrams. Added the AT commands list to set the codec in Slave-Burst (PCM) Mode configuration
1	2012-02-16	The present revision supersedes Rev. 0
0	2011-07-11	First issue

From Mod.0809 Rev.8

